PID Control Study – DMK Hydro Inc.

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CNG 633 SAAL

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Executive Summary

The study aims to design a proportional–integral–derivative (PID) temperature control system to regulate the temperature of the system using a heating element, specifically a heating tape, by measuring the system temperature with a thermocouple and adjusting the energy output through a solid-state relay (SSR). The construction of the PID temperature controller involves assembling various provided components, including the PID module, SSR, thermocouple, heating tape, wiring, and safety features like grounded wall plugs and fuses.

This report also includes a temperature control study, which involves operating the assembled PID controller to maintain a set temperature within specified limits while monitoring system performance at different flow rates. Quality requirements mandate temperature control at 25°C with set limits of ± 1% of the setpoint and a confidence level of 99.9%.

Additionally, a combined control study incorporating concentration control alongside temperature control is outlined. This involves operating both the PID temperature controller and the on-off PLC controller to regulate temperature and concentration simultaneously, with similar quality requirements for both variables.

The experiment's outcomes reveal that the PID control studies revealed consistent deviations from the setpoint temperature of 25.0°C at both flow rates, indicating inefficiencies in the control loop and potential issues with the thermocouple readings. Additionally, increased temperature measurements could result from the generated heat from the centrifugal pump. Meanwhile, the combined PID and PLC studies aimed to

regulate both temperature and dye concentration simultaneously. The PLC control demonstrated its ability to stabilize the system closer to the upper bound of the acceptable concentration range, despite deviations from the setpoint. However, both control strategies require further optimization to achieve control that is more precise and minimize errors.

In the future, enhancing the system's efficacy and optimization by changing PLC and PID parameters would be best.

Table of Contents

Executive Summary 2

Control Strategy – PID Control 5

Control Studies – PID Control 10

Control Studies – Combined Studies 13

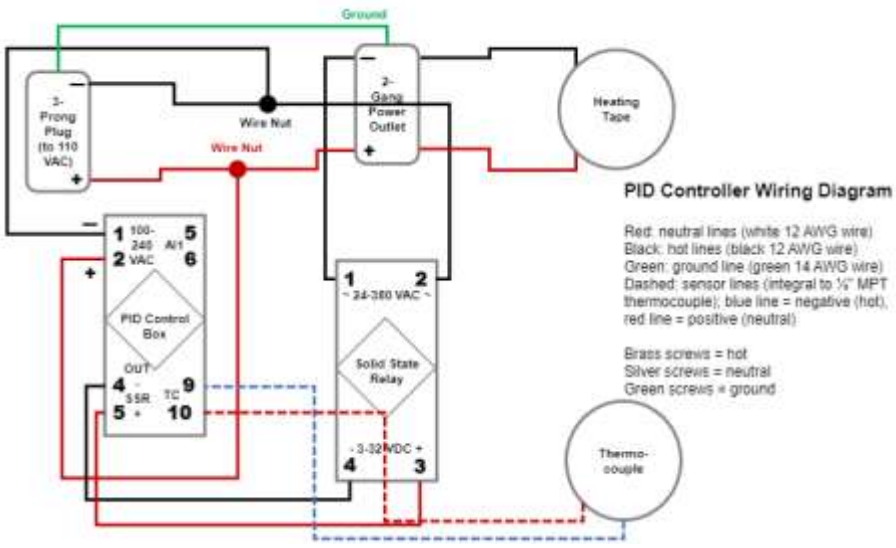
Appendix 19

Control Strategy – PID Control

Table 1: Overview of control loop characteristics for PID control

Control Loop Type	Proportional–Integral–Derivative (PID) Control
Subtype	Feedback Control
Measured Variable	Temperature
Manipulated Variable	Flowrate
Gain Type	Positive Gain
Potential Disturbances	The disturbance encountered by the plant involved a significant drop in the system's temperature, achieved by introducing a sudden decline using a bucket of ice.

Figure 1: Wiring diagram for the PID controller



Note. Wiring diagram for the PID controller. From "PID Temperature Controller Wiring Diagram.pdf" by George Clark, 2024. Copyright 2024 Seneca Polytechnic.

Figure 2: Annotated operational proportional–integral–derivative controller box on the pilot plant (1)

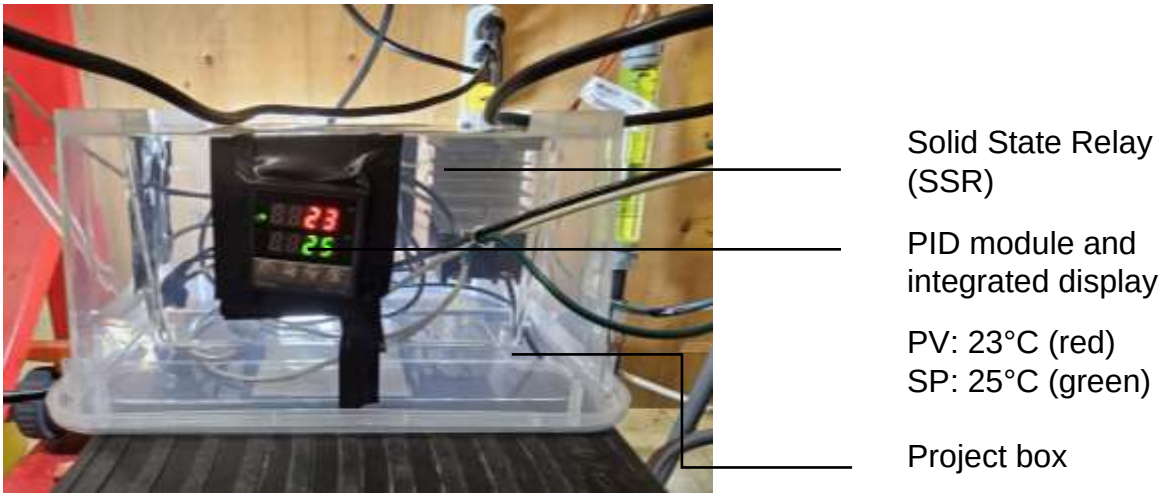


Figure 3: Annotated operational proportional–integral–derivative controller box on the pilot plant (2)

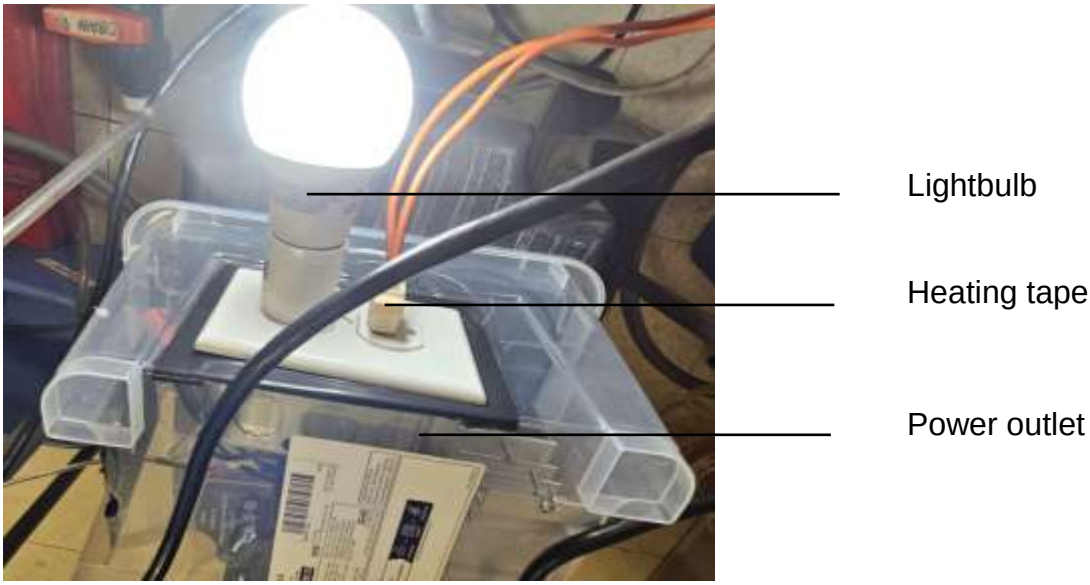


Figure 4: Annotated photograph of the programmable logic controller (PLC) on the pilot plant

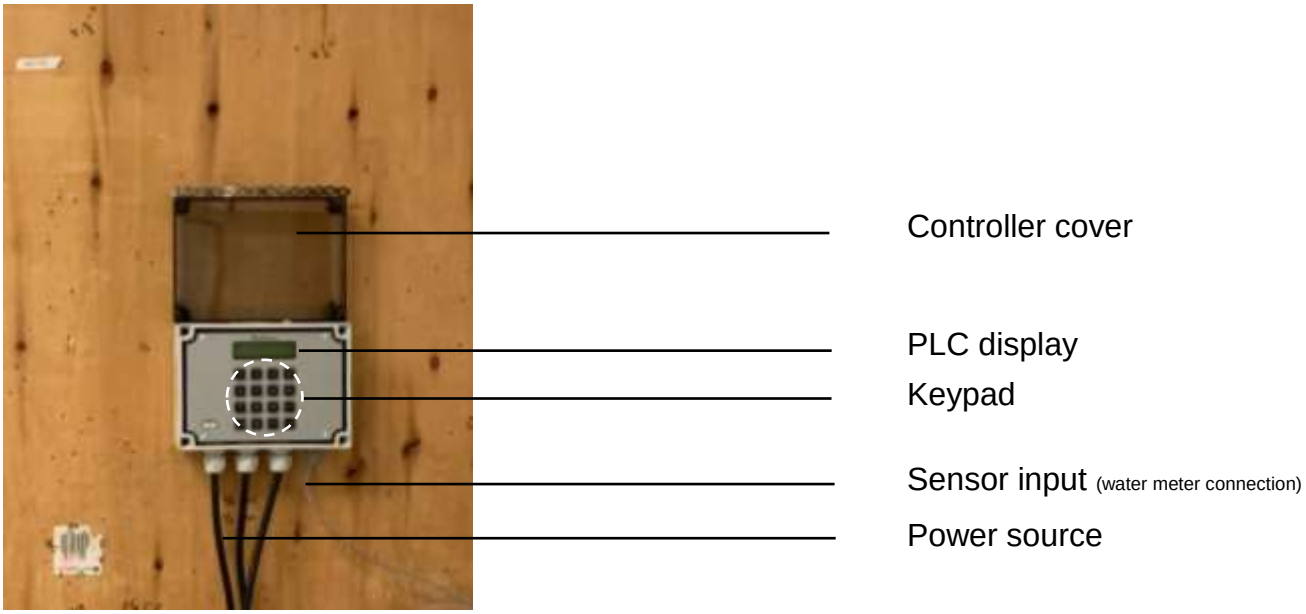


Figure 5: Annotated photograph of the water on the pilot plant

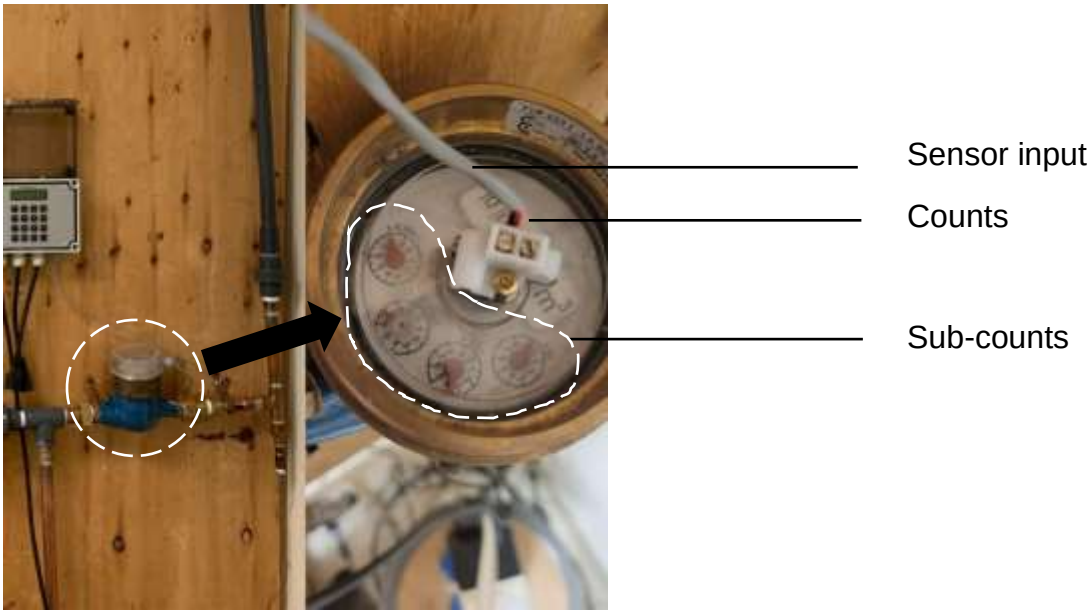


Figure 6: Annotated photograph of the diaphragm pumps on the pilot plant (1)

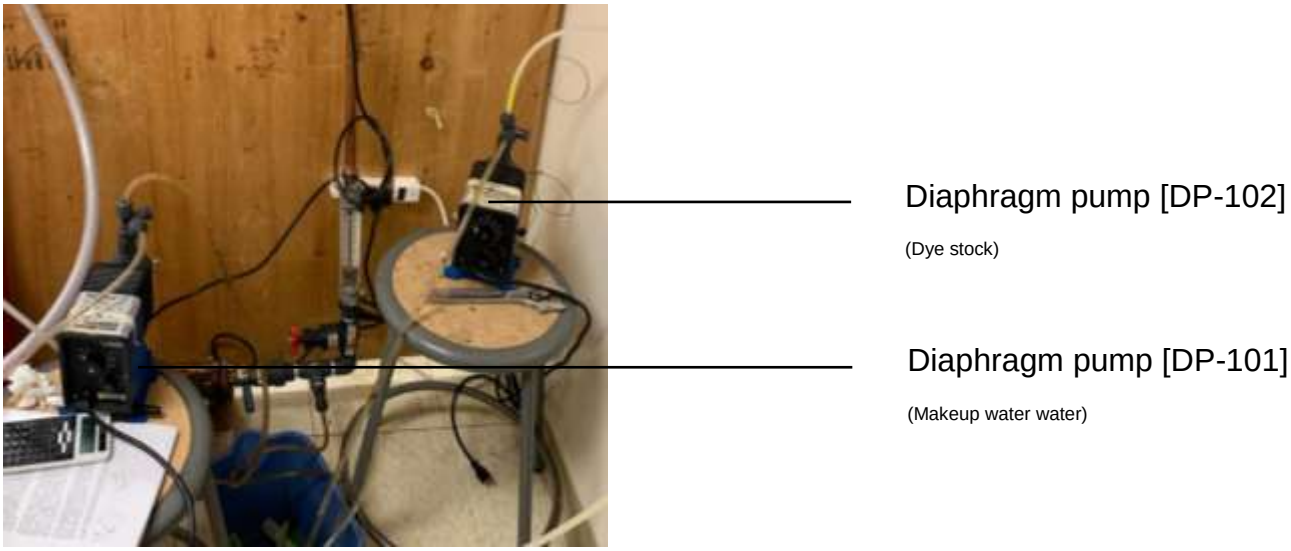


Figure 7: Annotated photograph of the diaphragm pumps on the pilot plant (2)

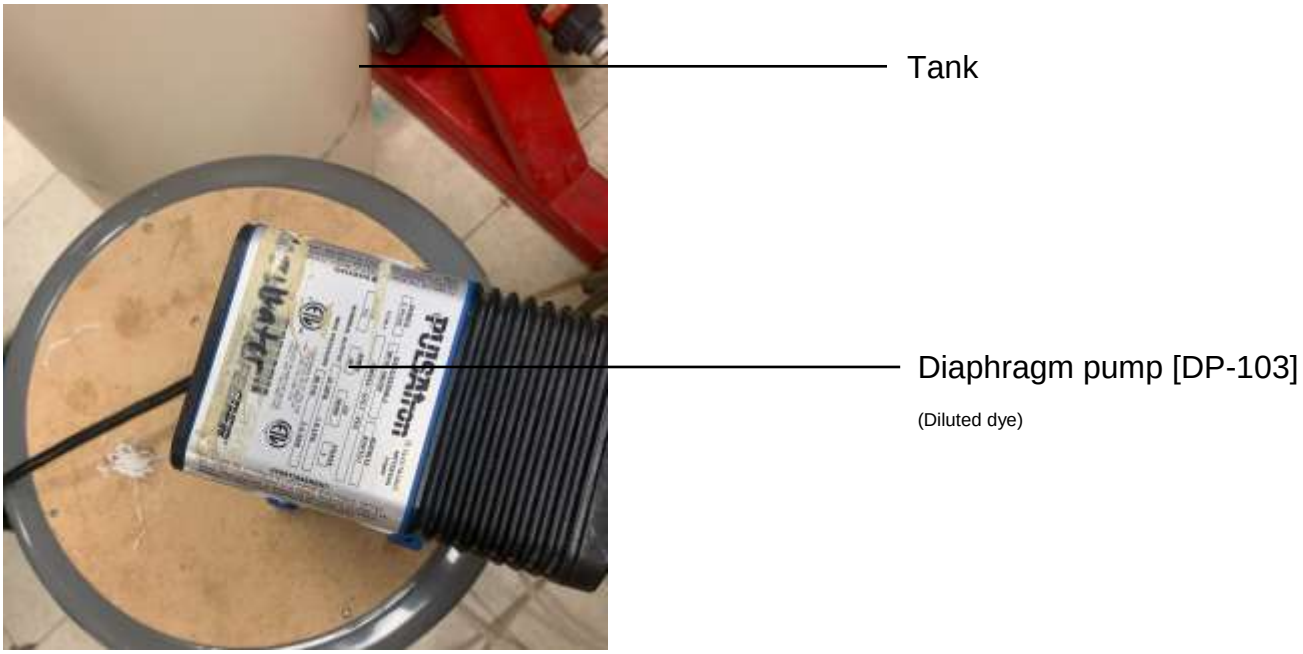


Figure 8: Annotated photograph of the heating tape on the pilot plant

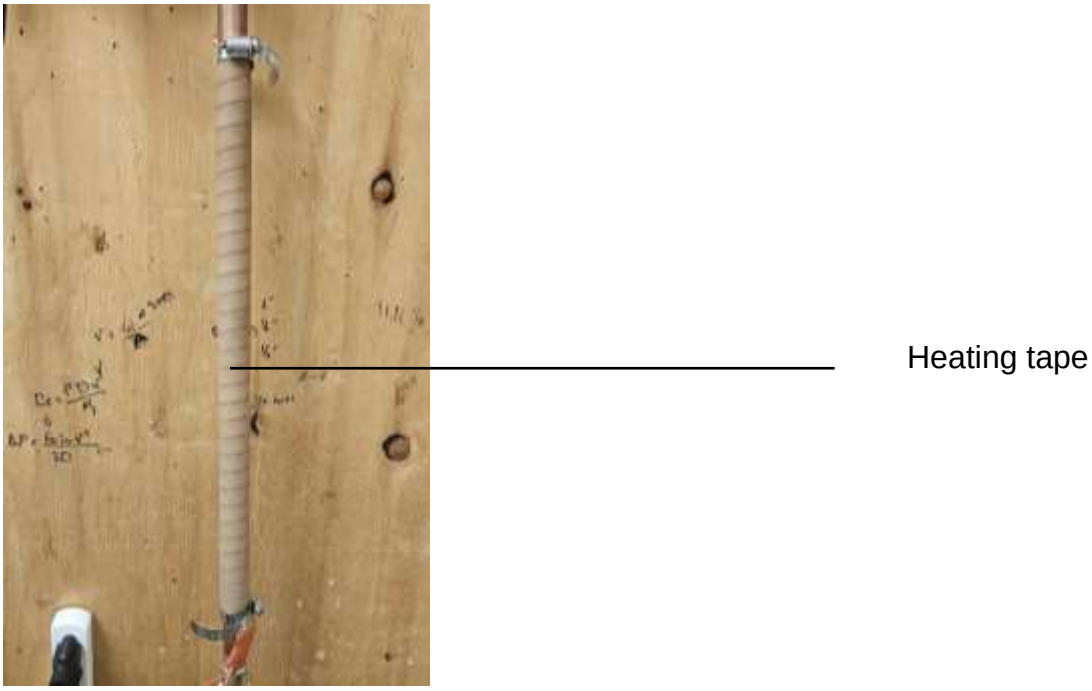


Figure 9: Annotated photograph of the entire pilot plant



Control Studies – PID Control

The operational condition or control status of the pilot plant with respect to meeting the setpoint of 25.0°C, displayed consistent deviations at 18.0 and 8.0 LPM, as shown in ***Figures 11*** and ***12*** with RSD values of approximately 1.7% and 1.6%, respectively. These deviations suggest that the PID feedback control loop is not effectively responding to changes in the system or external disturbances, leading to temperature fluctuations beyond the desired range. Two possible explanations could be because of a faulty thermocouple integrated into the project box and/or an increase in temperature because of the operation of the in-line centrifugal pump. The PID control loop operates by comparing the measured process variable to the desired setpoint to minimize any deviation however, the thermocouple consistently read approximately 2.0°C lower than the actual measurement. Consequently, the pump facilitated excessive heat transfer, leading to the observed upward trend in the process variable over time. Moreover, the system's inability to maintain control within a 30-minute interval resulted from heat dissipation from the copper piping housing the heating tape, in addition to the pilot plant's exposure to a vent. Additionally, both flowrates exhibit an upward trend over time in the process variable, leading to breaches of both the set point limits and the boundary limits at a 99.9% confidence interval. Therefore, the existing quality status of the system does not align with the desired setpoint of 25.0°C.

For the continued study of the pilot plant, with respect to its function, requires improving on the control and quality status through the modification of system conditions and equipment.

Figure 10: Represented data points on statistical process control (SPC) graphs



Where LSL and USL represent the upper and lower setpoint limits, and the UB and LB, represent the upper and lower bounds.

Figure 11: SPC (**99.9% CI**) – Temperature (°C) of dye delivered (min) at 18 LPM

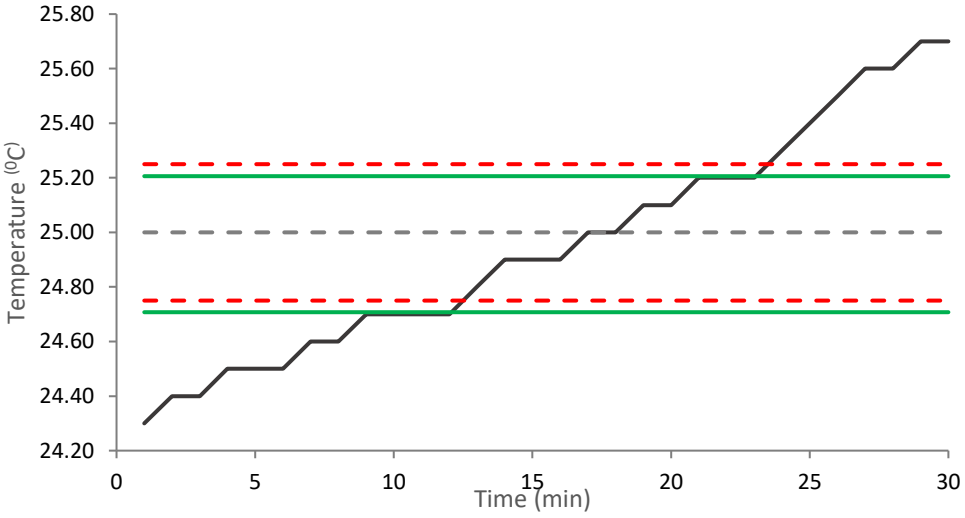
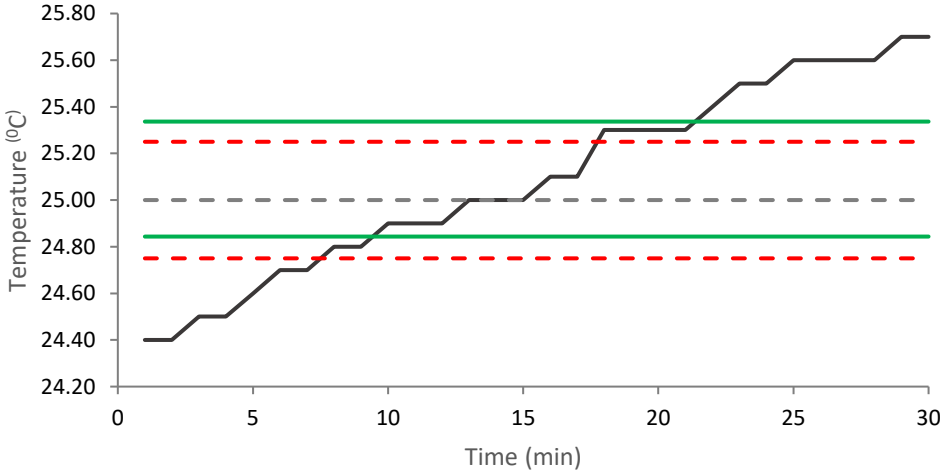


Figure 12: SPC (**99.9% CI**) – Temperature (°C) of dye delivered (min) at 8 LPM



The disturbance was a significant drop in temperature, achieved by introducing a sudden decline using a bucket of ice. The process variable over a 30-minute interval is shown in ***Figure 13***. From the plot, it can be inferred that the system attempted to enter the set point limits, doing this at a slow rate.

Figure 13: SPC (99.9% CI) – Temperature (°C) of dye delivered (min) for a 30-minute disturbance

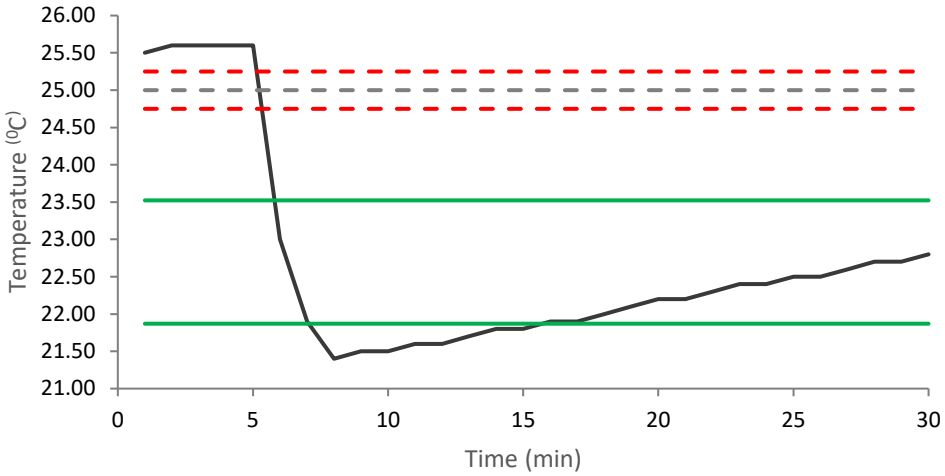


Table 2: Control of fluorescein dye concentration at various flowrates with 99.9 % confidence interval [*PID-1 (18.0 GPM), PID-2 (8.0 GPM), and PID-D (Disturbance)*]

	T_{sp} (°C)	$T_{avg,pv}$ (°C)	RSD (%)	Control Status	Quality Status
PID-1	25.0	24.8	1.7	Deviations from the SP indicate inefficiencies in the thermocouple attached to the PID control box, or heat transfer from the pump.	Upward trend over time in PV highlight breaches of both the SPLs and BLs at a 99.9% confidence interval.
PID-2		25.1	1.6		
PID-D		22.7	5.6	Deviations from the SP indicate a significant change in PV from a sudden decline.	System attempted to enter within the SPCs at a slow rate.

Control Studies – Combined Studies

The combined studies aims to incorporate the concentration control alongside temperature control. This involves operating both the PID temperature controller and the on-off PLC controller to regulate these variables simultaneously.

Table 3: Overview of control loop characteristics for PLC control

Control Loop Type	Programmable Logic Control (PLC)
Subtype	Feedforward Control
Measured Variable	Fluorescein dye concentration
Manipulated Variable	Operation of diaphragm pumps (flowrate and duration of pump inject)
Gain Type	N/a (binary system)

For the completion of this study, the operation of the pumps were adjusted to the conditions highlighted in ***Table 4***.

Table 4: Summary of the time (seconds) and frequency (counts) required to operate the dye injection pump and makeup water at their most precise flowrates

Dye pump	**Makeup water pump**
Most Precise [100/100]	
Time (s)	**Time (s)**
13	299
Frequency (counts)	**Frequency (counts)**
18	18

Sample calculation of time (s) and frequency (counts) of dye pump for the most precise flowrate [100/100]

Known variables

C_{in} = 200 ppm (dye stock concentration)

F_{out} = 0.386 mL/s (F_{avg} of diluted dye pump at most precise rate [50/100])

C_{out} = 4.88 ppm (midpoint from dye calibration plot at 485 nm)

$$\boldsymbol{F_{in}}C_{in} = F_{out}C_{out}$$

$$(F_{in})(200\ ppm) = (0.444\frac{mL}{s})(4.88\ ppm)$$

$$F_{in} = \frac{(0.386\frac{mL}{s})(4.88\ ppm)}{(200\ ppm)} = 0.0094\frac{mL}{s} = 0.57\ mL/min$$

Mass balance ratio

$$ratio = \frac{F_{theoretical}}{F_{measured}}$$

$$ratio = \frac{(0.57\frac{mL}{min})}{\left(0.444\frac{mL}{s} \times 60\ s\right)} = 0.0212$$

$$0.0212 \times \frac{60\ s}{1\ min} = \boldsymbol{1.3\frac{s}{min}}$$

Counts

F_{cal} = 18.3 LPM (F_{cal} rotameter)

$$counts\ per\ minute = \frac{18.3\frac{L}{min}}{10\frac{L}{count}} = \boldsymbol{1.8\frac{count}{min}}$$

Multiplier (factor of 10)

Time = 1.3 s/min = **13 seconds**

Count = 1.8 count/min = **18 counts**

The lower and upper control limits (LCL and UCL) provide the operational range within which the system should ideally function, reflecting its performance, while the upper and lower bounds at a 99.9 % confidence interval establishes a statistical range of acceptable concentrations based on the mean. These parameters are necessary in assessing the system's control status.

Figure 14: Represented data points on statistical process control (SPC) graphs



Figure 15: SPC **(99.9 % CI**) – Concentration (ppm) of the fluorescein dye delivered (min) for the most precise flowrate [100/100]

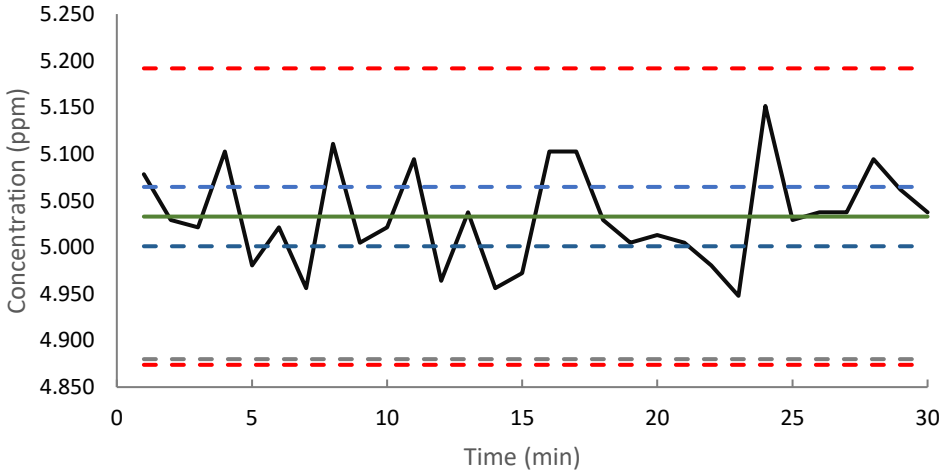
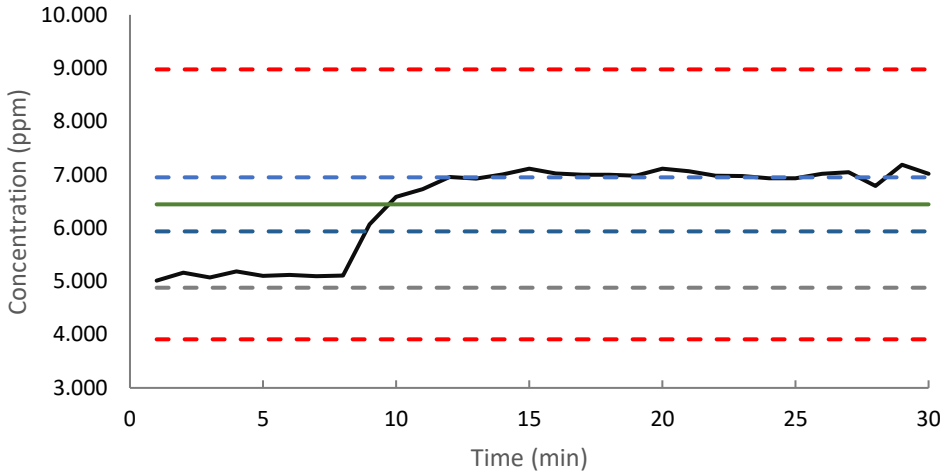


Figure 16: SPC **(99.9 % CI**) – Concentration (ppm) of the fluorescein dye delivered (min) for a 30-minute disturbance



The operational condition of the pilot plant with respect to maintaining the set point of 4.88 ppm, displayed a 1.1 % RSD with a mean concentration of 5.03 ± 0.05 ppm, as well as a 3.1 % error over consecutive 30-minute intervals, as shown in Figure 15. Figure 16 highlights the disturbance, with which the system displayed a mean concentration of 6.44 ± 0.84 ppm, a 13.10 % RSD and a 32.05 % error from the original set point. After the disturbance, the system stabilized around a point as a method of trying to fix itself instead of reverting to its original set point of 4.88 ppm. This behaviour is a characteristic of an on/off PLC system, achieving success by stabilizing closer to the upper bound of the 99.9 % CI. In both instances, the system is/regains control.

Refer to ***Figure 10*** for the statistical processing of ***Figure 17*** and ***18***.

Figure 17: SPC (**99.9% CI**) – Temperature (°C) of dye delivered (min) for a 30-minute interval

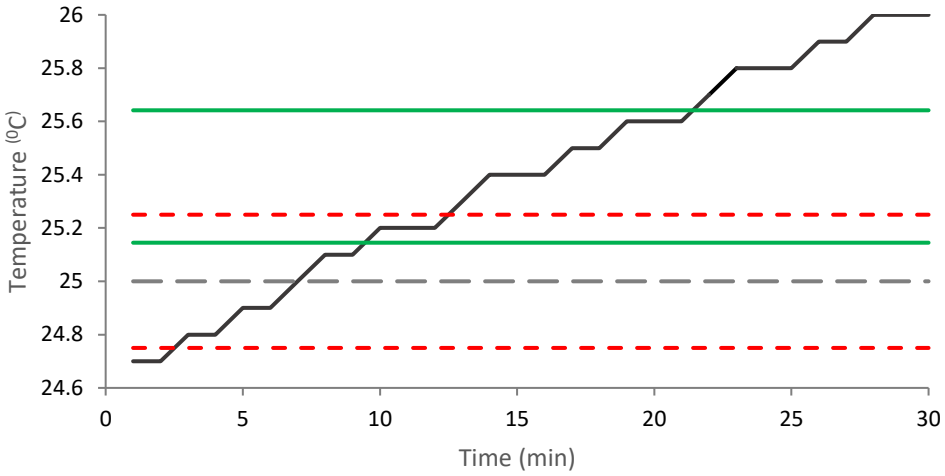
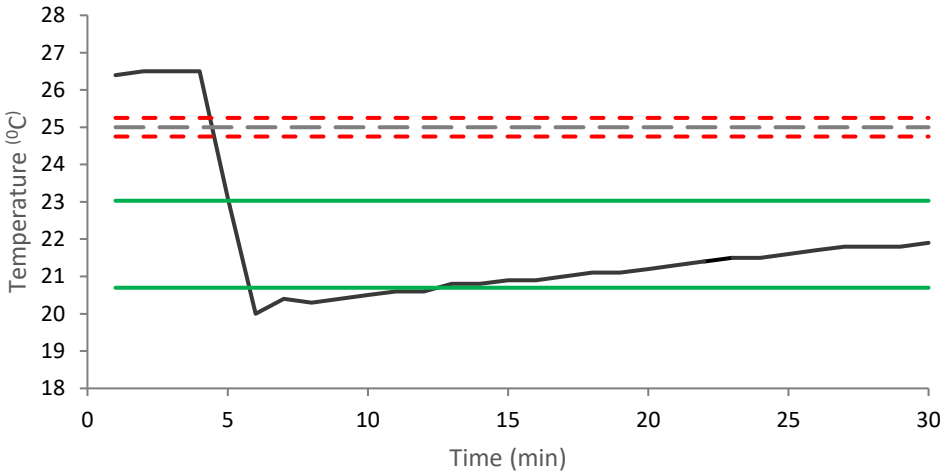


Figure 18: SPC (**99.9% CI**) – Temperature (°C) of dye delivered (min) for a 30-minute disturbance



The operational condition of the pilot plant with respect to maintaining a set point of 25.0°C shows similar results in ***Figure 17*** as ***Figures 11*** and ***12***. The mean temperature was 25.4 ± 0.4°C with a 1.6 % RSD and % error. ***Figure 18*** highlights an attempt made by the

PID controller to stabilize the significant drop in temperature caused by a bucket of ice. The temperature, in comparison, increases at a much slower rate. With a mean temperature of 21.9 ± 1.9°C and 8.9 % RSD. In both instances, the system was not in control as they fell outside the SPCs. It is important to note that the thermocouple measuring the process variable within the tank reported approximately 2°C lower than the true temperature. This was verified using other temperature probes. This source of error could contribute to the increasing trend in temperature over 30 minutes as shown in ***Figure 17***, without achieving stabilization.

Appendix

Table 5: Temperature (°C) measurements for fluorescein dye for PID study (18 LPM)

Time (min)	T (°C)	T_{avg} (ppm)	Std. Dev. (±)	LSL	USL	99.9% Error (ppm)	Upper Bound (ppm)	Lower Bound (ppm)
1	24.300	24.957	0.415	24.750	25.250	0.249	25.206	24.707
2	24.400							
3	24.400							
4	24.500							
5	24.500							
6	24.500							
7	24.600							
8	24.600							
9	24.700							
10	24.700							
11	24.700							
12	24.700							
13	24.800							
14	24.900							
15	24.900							
16	24.900							
17	25.000							
18	25.000							
19	25.100							
20	25.100							
21	25.200							
22	25.200							
23	25.200							
24	25.300							
25	25.400							
26	25.500							
27	25.600							
28	25.600							
29	25.700							
30	25.700							

Table 6: Temperature (°C) measurements for fluorescein dye for PID study (8 LPM)

Time (min)	T (°C)	T_{avg} (ppm)	Std. Dev. (±)	LSL	USL	99.9% Error (ppm)	Upper Bound (ppm)	Lower Bound (ppm)
1	24.400	25.090	0.411	24.750	25.250	0.247	25.337	24.843
2	24.400							
3	24.500							
4	24.500							
5	24.600							
6	24.700							
7	24.700							
8	24.800							
9	24.800							
10	24.900							
11	24.900							
12	24.900							
13	25.000							
14	25.000							
15	25.000							
16	25.100							
17	25.100							
18	25.300							
19	25.300							
20	25.300							
21	25.300							
22	25.400							
23	25.500							
24	25.500							
25	25.600							
26	25.600							
27	25.600							
28	25.600							
29	25.700							
30	25.700							

Table 7: Temperature (°C) measurements for fluorescein dye for combined study (18 LPM)

Time (min)	T (°C)	T_{avg} (ppm)	Std. Dev. (±)	LSL	USL	99.9% Error (ppm)	Upper Bound (ppm)	Lower Bound (ppm)
1	24.700	25.393	0.414	24.750	25.250	0.248	25.642	24.145
2	24.700							
3	24.800							
4	24.800							
5	24.900							
6	24.900							
7	25.000							
8	25.100							
9	25.100							
10	25.200							
11	25.200							
12	25.200							
13	25.300							
14	25.400							
15	25.400							
16	25.400							
17	25.500							
18	25.500							
19	25.600							
20	25.600							
21	25.600							
22	25.700							
23	25.800							
24	25.800							
25	25.800							
26	25.900							
27	25.900							
28	26.000							
29	26.000							
30	26.000							

Table 8: Temperature (°C) measurements for fluorescein dye for combined study (18 LPM)

Time (min)	T (°C)	T_{avg} (ppm)	Std. Dev. (±)	LSL	USL	99.9% Error (ppm)	Upper Bound (ppm)	Lower Bound (ppm)
1	26.400	21.863	1.941	24.750	25.250	1.166	23.029	20.697
2	26.500							
3	26.500							
4	26.500							
5	23.100							
6	20.000							
7	20.400							
8	20.300							
9	20.400							
10	20.500							
11	20.600							
12	20.600							
13	20.800							
14	20.800							
15	20.900							
16	20.900							
17	21.000							
18	21.100							
19	21.100							
20	21.200							
21	21.300							
22	21.400							
23	21.500							
24	21.500							
25	21.600							
26	21.700							
27	21.800							
28	21.800							
29	21.800							
30	21.900							

Table 9: Concentration (ppm) measurements for fluorescein dye for combined study

Time (min)	Abs (a.u.)	Concentration (ppm)	C_{avg} (ppm)	Std. Dev. (±)	LCL	UCL	99.9% Error (ppm)	Upper Bound (ppm)	Lower Bound (ppm)
1	0.628	5.078	5.033	0.053	4.831	4.929	0.032	5.065	5.001
2	0.622	5.029							
3	0.621	5.021							
4	0.631	5.103							
5	0.616	4.980							
6	0.621	5.021							
7	0.613	4.956							
8	0.632	5.111							
9	0.619	5.005							
10	0.621	5.021							
11	0.630	5.094							
12	0.614	4.964							
13	0.623	5.037							
14	0.613	4.956							
15	0.615	4.972							
16	0.631	5.103							
17	0.631	5.103							
18	0.622	5.029							
19	0.619	5.005							
20	0.62	5.013							
21	0.619	5.005							
22	0.616	4.980							
23	0.612	4.948							
24	0.637	5.151							
25	0.622	5.029							
26	0.623	5.037							
27	0.623	5.037							
28	0.630	5.094							
29	0.626	5.062							
30	0.623	5.037							

Table 10: Concentration (ppm) measurements for fluorescein dye for combined study

Time (min)	Abs (a.u.)	Concentration (ppm)	C_{avg} (ppm)	Std. Dev. (±)	LCL	UCL	99.9% Error (ppm)	Upper Bound (ppm)	Lower Bound (ppm)
1	0.620	5.013	4.880	0.844	4.831	4.929	0.507	6.951	5.937
2	0.638	5.160							
3	0.627	5.070							
4	0.641	5.184							
5	0.631	5.103							
6	0.633	5.119							
7	0.630	5.094							
8	0.632	5.111							
9	0.750	6.072							
10	0.813	6.585							
11	0.831	6.731							
12	0.859	6.959							
13	0.855	6.927							
14	0.865	7.008							
15	0.878	7.114							
16	0.867	7.024							
17	0.864	7.000							
18	0.864	7.000							
19	0.862	6.984							
20	0.878	7.114							
21	0.872	7.065							
22	0.862	6.984							
23	0.861	6.976							
24	0.856	6.935							
25	0.856	6.935							
26	0.866	7.016							
27	0.870	7.049							
28	0.838	6.788							
29	0.887	7.187							
30	0.866	7.016							